AMERICAN UNIVERSITY OF BEIRUT
MAROUN SEMAAN FACULTY OF ENGINEERING & ARCHITECTURE

Android Application Development

EECE 451 – Mobile Networks and Applications

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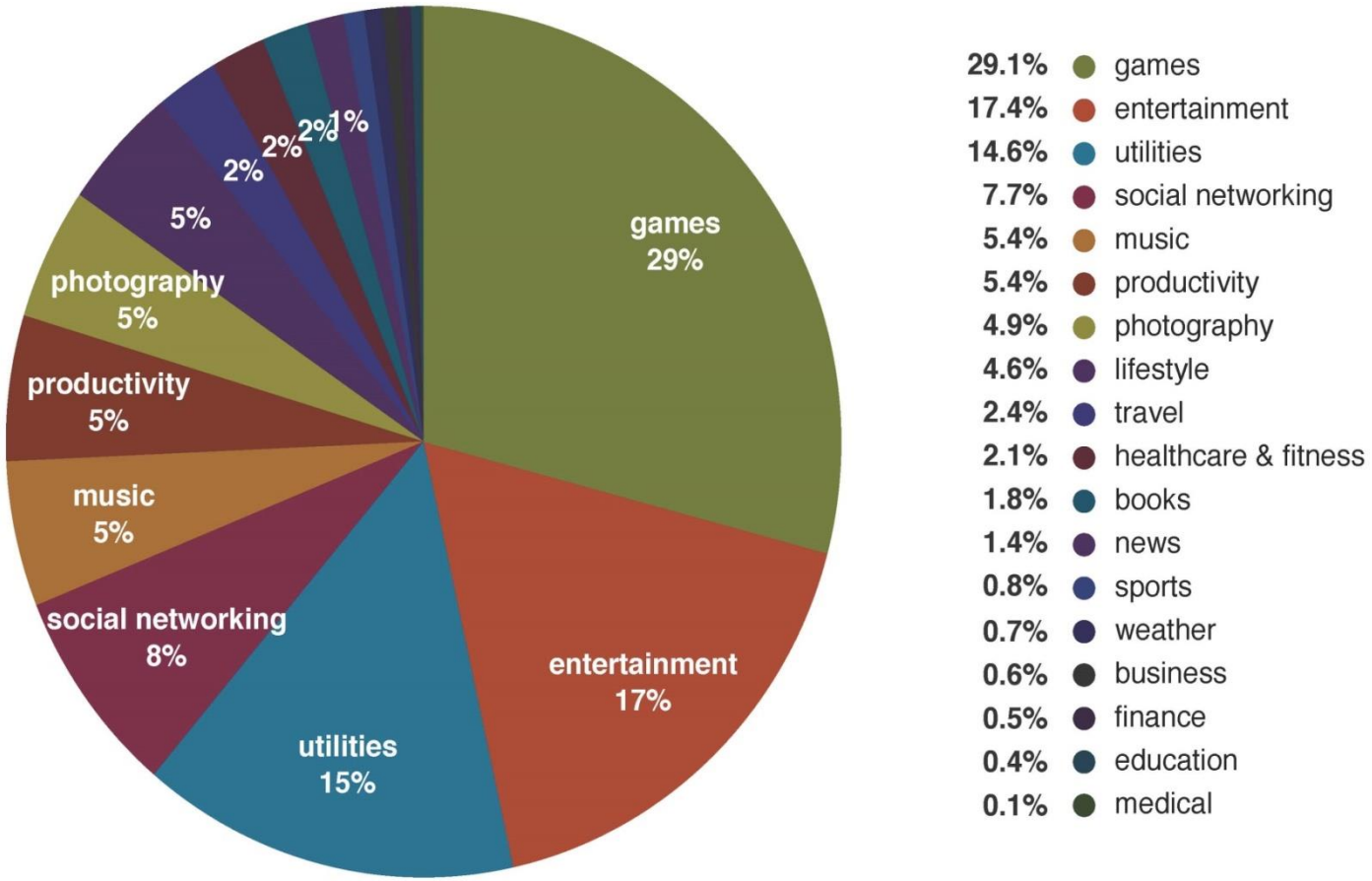
What is Android?

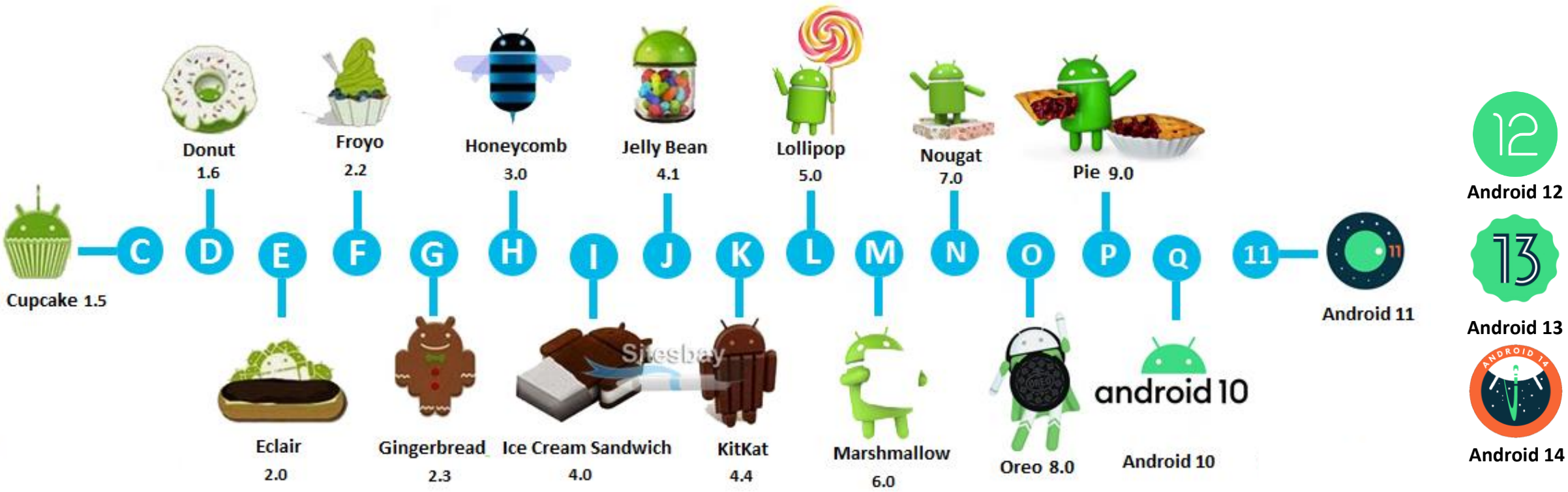
- Open source OS designed for mobile devices.
- Developed by Google
- Based on Linux
- Features of Android
  - Beautiful UI, Connectivity, Storage, Media support, Messaging, Web browser, Multi-touch, Multi-tasking, Resizable widgets, Multi-Language, GCM, Wi-Fi Direct, Android Beam
- Android Applications
  - Mostly developed with Java
  - Other languages could be used

Categories of Android applications



Android versions

- Versions range from A to P, then numbers.
- The current latest version is Android 14 (released October 2023)

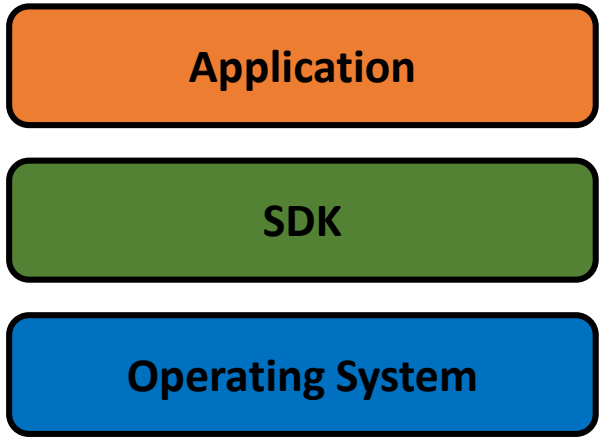


https://www.computerworld.com/article/3235946/android-versions-a-living-history-from-1-0-to-today.html?page=1

Android SDK

- Each android version has a corresponding Software Development Kit (SDK)
- SDK version is denoted by its API level
- The SDK provides the necessary APIs (Application Programming Interfaces) and tools to develop applications that can run on various Android versions and devices.

ANDROID PLATFORM VERSION		API LEVEL
4.0	Ice Cream Sandwich	15
4.1	Jelly Bean	16
4.2	Jelly Bean	17
4.3	Jelly Bean	18
4.4	KitKat	19
5.0	Lollipop	21
5.1	Lollipop	22
6.0	Marshmallow	23
7.0	Nougat	24
7.1	Nougat	25
8.0	Oreo	26
8.1	Oreo	27
9.0	Pie	28
10.	Android 10	29



Android development versions

- When developing, don't always target the newest version!

Android Platform/API Version Distribution

ANDROID PLATFORM VERSION	API LEVEL	CUMULATIVE DISTRIBUTION
4.0 Ice Cream Sandwich	15	
4.1 Jelly Bean	16	99.8%
4.2 Jelly Bean	17	99.2%
4.3 Jelly Bean	18	98.4%
4.4 KitKat	19	98.1%
5.0 Lollipop	21	94.1%
5.1 Lollipop	22	92.3%
6.0 Marshmallow	23	84.9%
7.0 Nougat	24	73.7%
7.1 Nougat	25	66.2%
8.0 Oreo	26	60.8%
8.1 Oreo	27	53.5%
9.0 Pie	28	39.5%
10. Android 10	29	8.2%

The minimum SDK version determines the lowest level of Android that your app will run on.

You typically want to target as many users as possible, so you would ideally want to support everyone -- with a minimum SDK version of 1. However, that has some disadvantages, such as lack of features, and very few people use devices that old anymore.

Your choice of minimum SDK level should be a tradeoff between the distribution of users you wish to target and the features that your application will need.

Click each Android Version/API level for more information.

OK Cancel

Android IDEs

- There are several options
- Android Studio is the most recommended

Android Studio

Eclipse

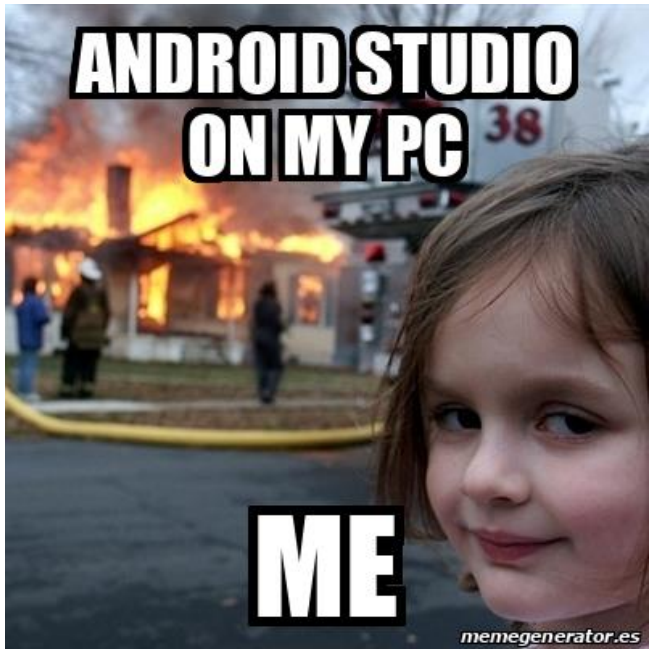
Visual Studio

NetBeans

Requirements for Android Studio

The following are the minimum system requirements for Android Studio

- ✓ 8 GB RAM or more
- ✓ 8 GB of available disk space minimum (IDE + Android SDK + Android Emulator)
- ✓ 1280 x 800 minimum screen resolution



Installation

Java

1. Visit http://www.oracle.com/technetwork/java/javase/downloads/index.html
2. Choose your operating system
3. Install the corresponding JDK
4. Set the JAVA_HOME environment variable: search for Environment Variables and add a new system variable JAVA_HOME that points to your JDK folder, for example C:\Program Files\Java\jdk1.7.0_45

Online Tutorial: https://www.youtube.com/watch?v=jPwrWjEwtrw

Installation

Android Studio

1. Visit http://developer.android.com/sdk/index.html
2. Click the button *Download Android Studio*.
3. Accept terms, and click *Download*.
4. Run executable file of setup.
5. Follow the setup wizard to install Android Studio and any necessary SDK tools.

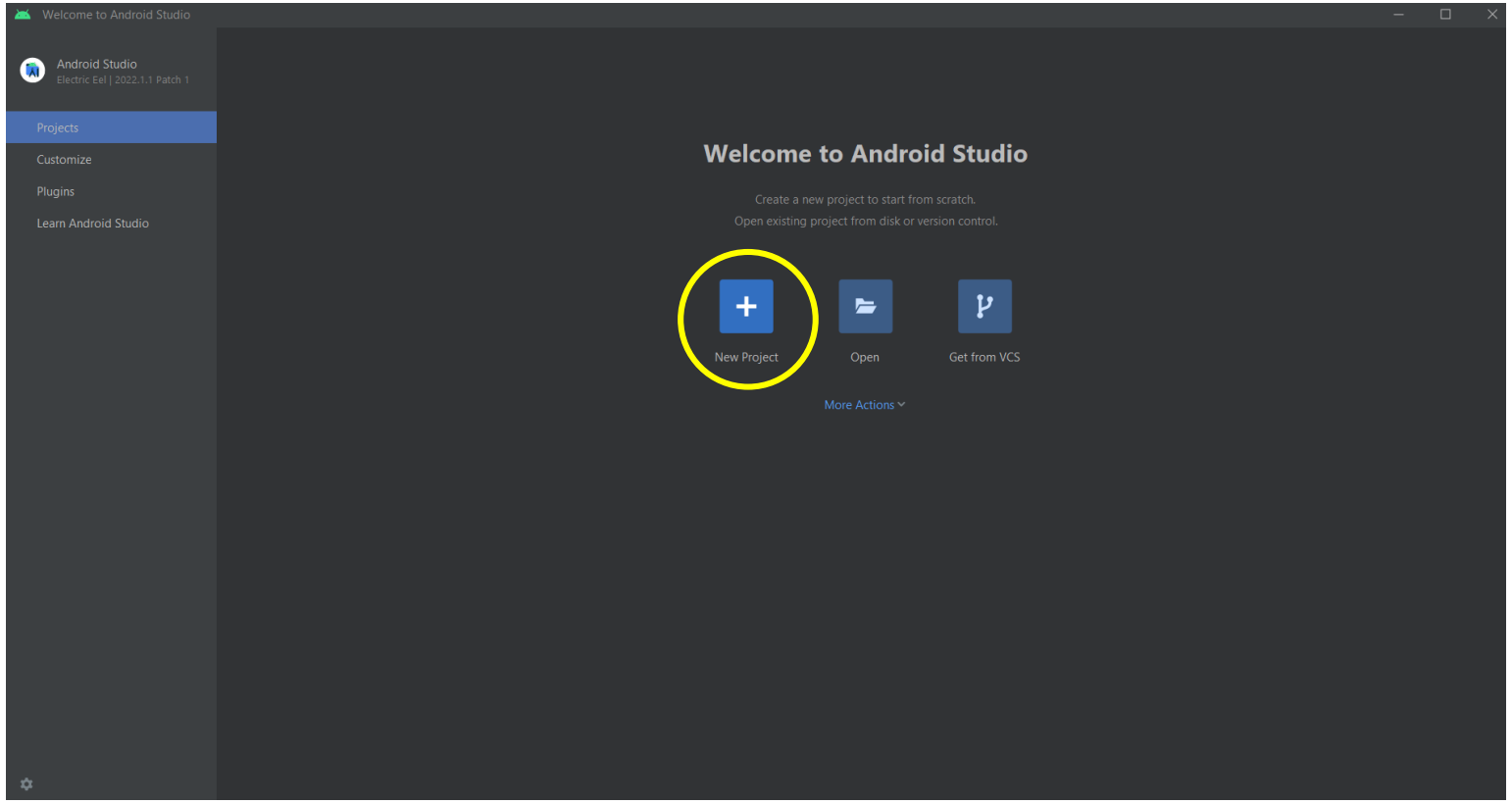
Online tutorial: https://www.youtube.com/watch?v=tNkgVd13ASI&t=605s

Building our First Application

With Android Studio

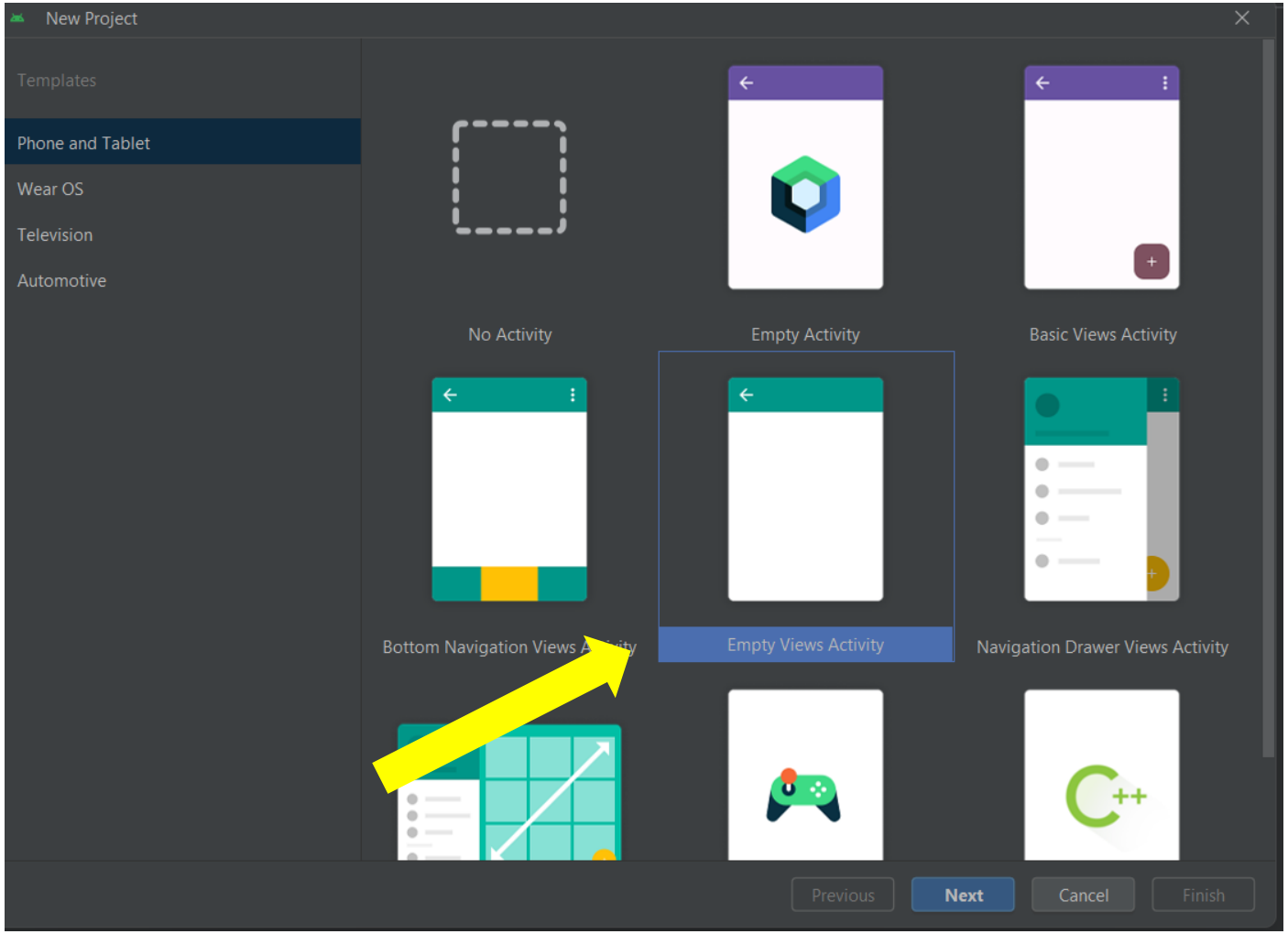
Step 1: Creating a new project

- Open Android Studio
- Click on New Project



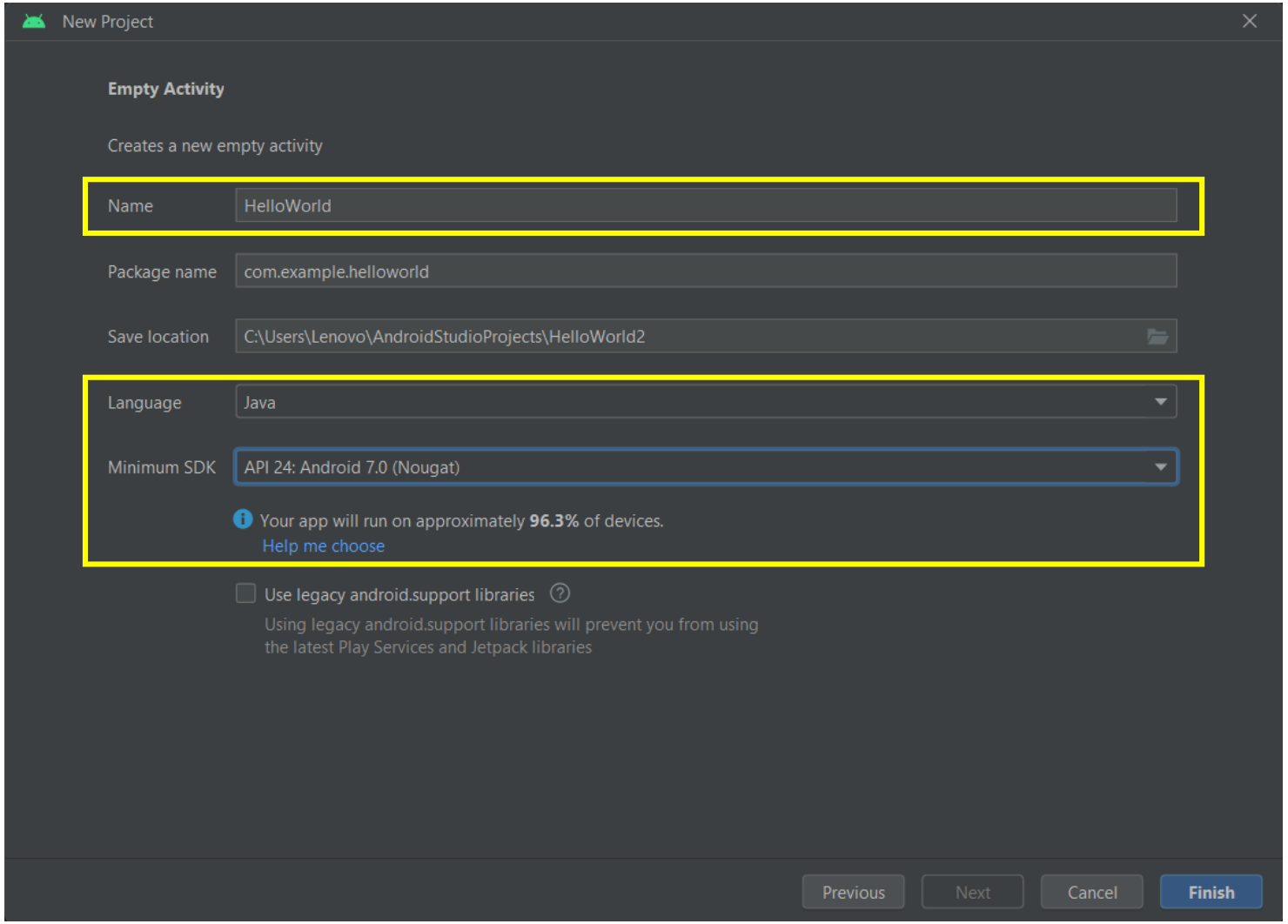
Step 1: Creating a new project

- Select “Empty Views Activity”
- Click “Next”



Step 1: Creating a new project

- Choose project name
- Specify language
- Specify minimum SDK
- Click “Finish”



Step 2: Inspecting project files

File Edit View Navigate Code Refactor Build Run Tools VCS Window Help HelloWorld - MainActivity.java [HelloWorld.app.main]

HelloWorld2 〉 app 〉 src 〉 main 〉 java 〉 com 〉 example 〉 helloworld 〉 MainActivity

app | Pixel 2 API 24

Project

- HelloWorld2 [HelloWorld] C:\Users\Lenovo\AndroidStudioProj
  - .gradle
  - .idea
  - app
    - libs
    - src
      - androidTest
      - main
        - java
          - com.example.helloworld
            - MainActivity
        - res
          - drawable
          - drawable-v24
          - layout
            - activity_main.xml
          - mipmap-anydpi-v26
          - mipmap-anydpi-v33
          - mipmap-hdpi
          - mipmap-mdpi
          - mipmap-xhdpi
          - mipmap-xxhdpi
          - mipmap-xxxhdpi
          - values
          - values-night
          - xml
        - AndroidManifest.xml
      - test [unitTest]
    - .gitignore
    - build.gradle
    - proguard-rules.pro
  - gradle
  - .gitignore
  - build.gradle

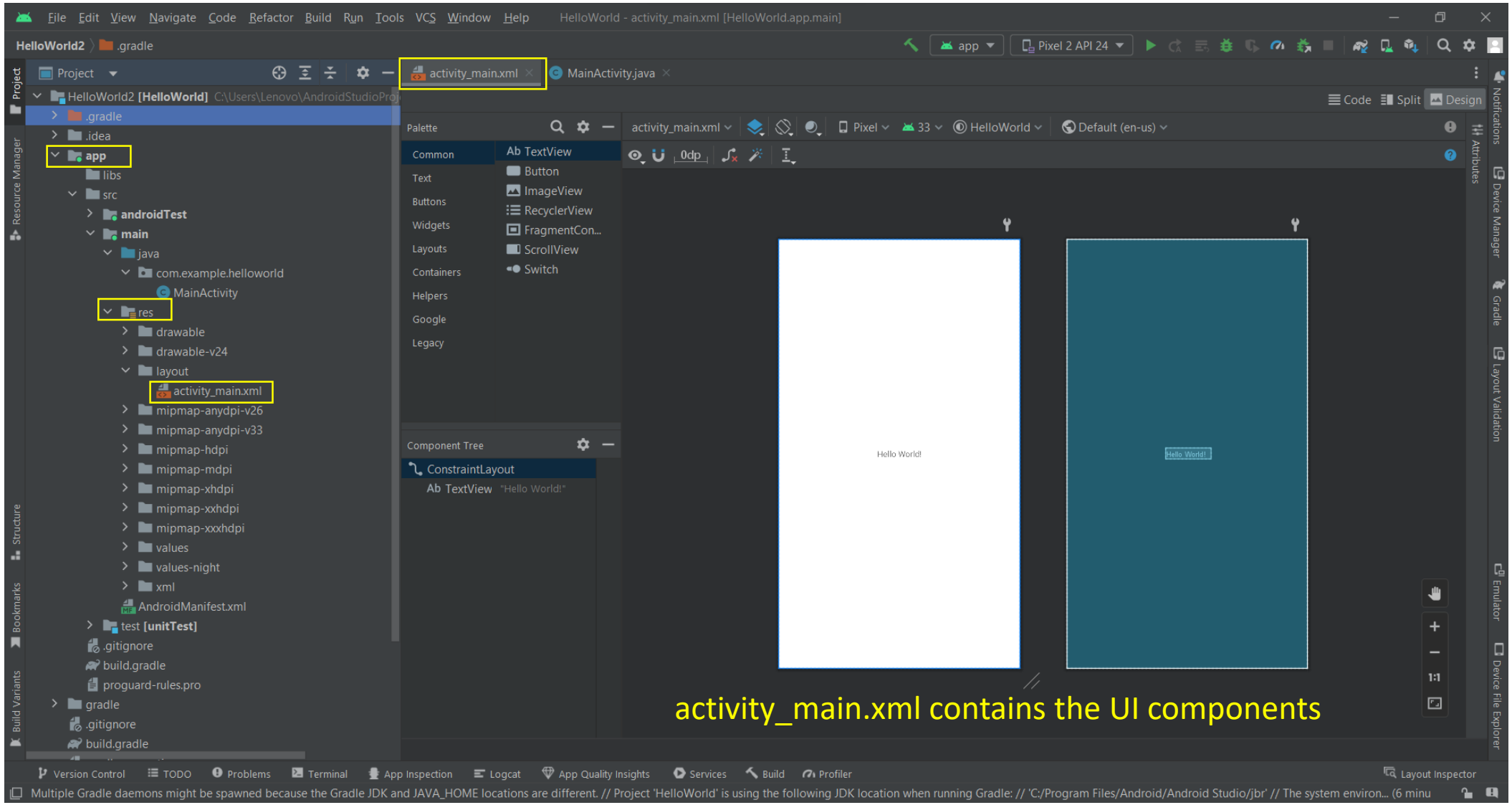
activity_main.xml | MainActivity.java

```
package com.example.helloworld;

import ...

2 usages
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

MainActivity.java contains the core code that executes once the application runs

Version Control | TODO | Problems | Terminal | App Inspection | Logcat | App Quality Insights | Services | Build | Profiler | Layout Inspector

Step 2: Inspecting project files



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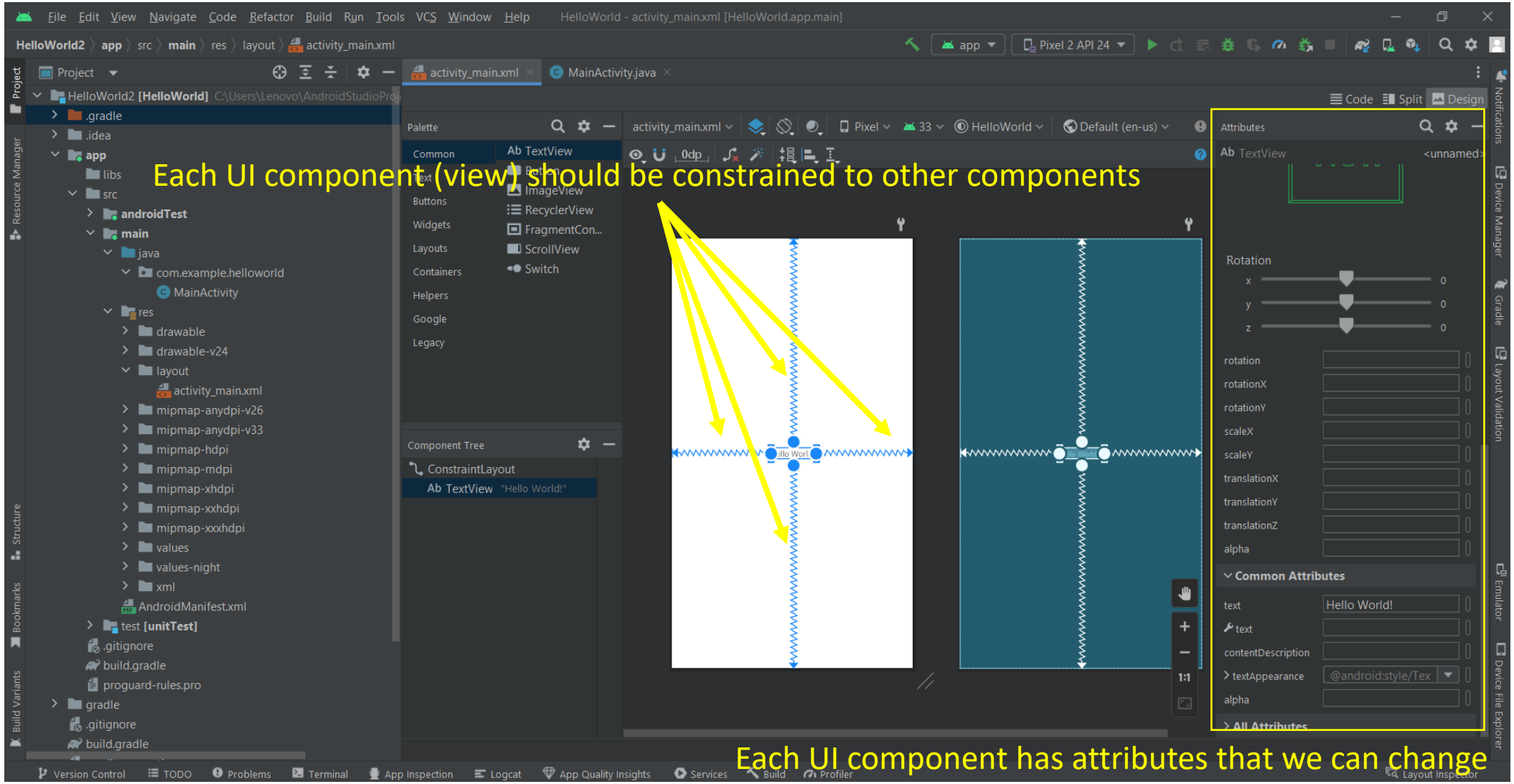
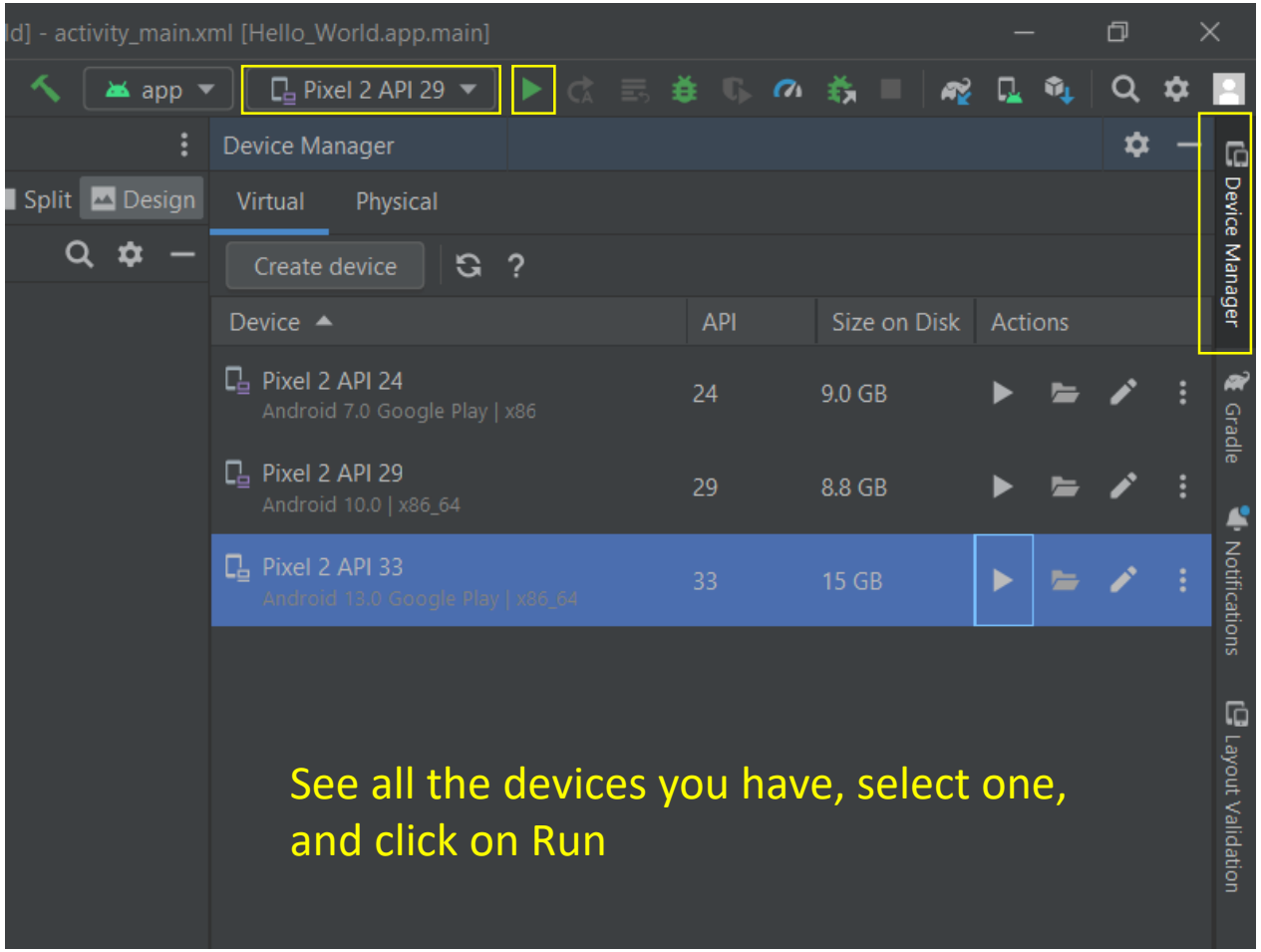


Step 3: Running the App

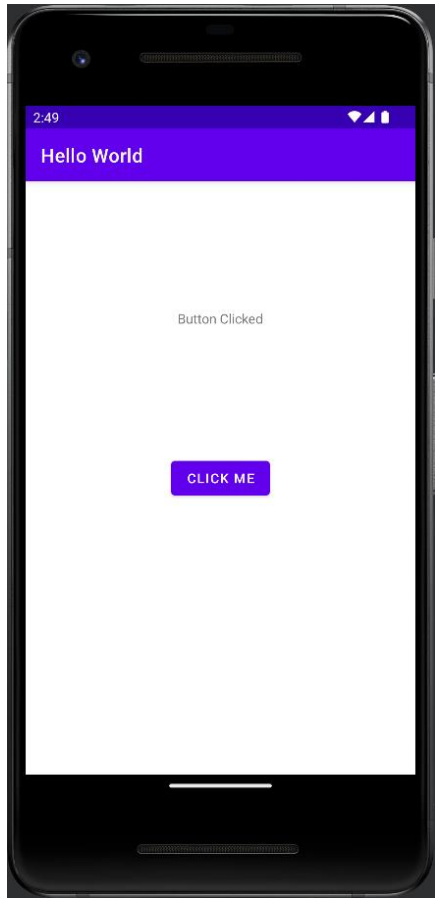


Step 4: Update App functionality

- Add a button and a text view to the UI
- Write the following code in the main activity
- Run and see the changes

```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button button = (Button) findViewById(R.id.button);
        button.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                TextView textview = (TextView) findViewById(R.id.textView);
                textview.setText("Button Clicked");
            }
        });
    }
}
```

Finding a view by ID